Solution to Exercise Sheet 5:
Empirical distributions and learning

Foundations of Neuro-Symbolic AI
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Task 1:
The empirical distribution $\mathbb{P}^D[X_0, X_1, X_2]$ is the average of the one-hot encodings of each observed state tuple.

(i) We introduce a decomposition variable I taking states in $[10]$ enumerating the days. Based on this variable we find a CP decompositon

$$\mathbb{P}^D[X_0, X_1, X_2] = \left\langle \mathbb{P}[X_0|I], \mathbb{P}[X_1|I], \mathbb{P}[X_1|I], \mathbb{P}[I] \right\rangle_{[X_0,X_1,X_2]}$$

where $\mathbb{P}[I] = \frac{1}{10}\mathbb{I}[I]$ and the leg matrices are

$$\mathbb{P}[X_0|I] = \begin{pmatrix} 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 \end{pmatrix}[X_0, I]$$
$$\mathbb{P}[X_1|I] = \begin{pmatrix} 1 & 1 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 & 1 & 1 \end{pmatrix}[X_1, I]$$
$$\mathbb{P}[X_2|I] = \begin{pmatrix} 1 & 0 & 1 & 0 & 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 & 0 & 0 & 1 & 1 \end{pmatrix}[X_1, I]$$

(ii) We want to find the contraction of $\mathbb{P}^D[X_0, X_1, X_2]$ with the propositional formula

$$f[X_1, X_2] = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}[X_1, X_2]\,,$$

representing the event that either both Alice and Bob bring and umbrella or both do not bring an umbrella. The rate of days on whis this is true is the contraction

$$\left\langle \mathbb{P}^D[X_0, X_1, X_2], f[X_1, X_2] \right\rangle_{[\varnothing]}$$

We use the CP decomposition above and can ignore the core $\mathbb{P}[X_0|I]$ since X_0 (Rain) does not appear. Further, we notice, that $\mathbb{P}[I]$ only brings a factor of $\frac{1}{10}$. We are left with computing

$$\frac{1}{10} \cdot \left\langle \mathbb{P}[X_1|I], \mathbb{P}[X_1|I], f[X_1, X_2] \right\rangle_{[\varnothing]} = \frac{5}{10} = \frac{1}{2}\,.$$

Task 2:
We investigate whether the three conditional independencies hold:

(i) $(X_1 \perp X_2) \mid X_0$: We have

$$\mathbb{P}^D[X_1, X_2|X_0 = 0] = \frac{1}{4} \cdot \mathbb{I}[X_1, X_2] = \frac{1}{2}\mathbb{I}[X_1] \otimes \frac{1}{2}\mathbb{I}[X_2] \quad \text{and}$$
$$\mathbb{P}^D[X_1, X_2|X_0 = 1] = \frac{1}{6}\begin{pmatrix} 1 & 1 \\ 2 & 2 \end{pmatrix}[X_1, X_2] = \frac{1}{3}\begin{pmatrix} 1 \\ 2 \end{pmatrix}[X_1] \otimes \frac{1}{2}\mathbb{I}[X_2]$$

and thus

$$(X_1 \perp X_2) \mid X_0\,.$$

This means that Alice and Bob bring umbrella indenpendently from each other given whether it rains.

(ii) $(X_0 \perp X_2) \mid X_1$: We find the same conditional distributions as in (i) (when interchanging variables):

$$\mathbb{P}^D\left[X_0, X_2 | X_1 = 0\right] = \frac{1}{4} \cdot \mathbb{I}\left[X_0, X_2\right] \quad \text{and}$$
$$\mathbb{P}^D\left[X_1, X_2 | X_1 = 1\right] = \frac{1}{6}\begin{pmatrix} 1 & 1 \\ 2 & 2 \end{pmatrix}\left[X_0, X_2\right].$$

Therefore, also $(X_0 \perp X_2) \mid X_1$ holds.

(iii) $(X_0 \perp X_1) \mid X_2$: We have

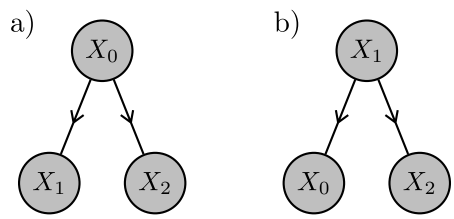
$$\mathbb{P}^D\left[X_0, X_1 | X_2 = 0\right] = \frac{1}{5}\begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}\left[X_0, X_1\right].$$

This is not the tensor product of two vectors, therefore $(X_0 \perp X_1) \mid X_2$ does not hold.

Task 3:

Notice, that in the unregularized learning setup, the solution is the empirical distribution itself represented as a Bayesian network.

(i) There are two directed hypergraphs capturing both exactly one of the two conditional independence statements:



To be more precise, hypergraph $a)$ captures $(X_1 \perp X_2) \mid X_0$ and hypergraph $b)$ captures $(X_0 \perp X_2) \mid X_1$.

(ii) Based on the calculations in task 2 the optimal decorating conditional distributions are in the case $a)$

$$\mathbb{P}^D\left[X_1 | X_0\right] = \begin{pmatrix} \frac{1}{2} & \frac{1}{3} \\ \frac{1}{2} & \frac{2}{3} \end{pmatrix}\left[X_1, X_0\right]$$
$$\mathbb{P}^D\left[X_2 | X_0\right] = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}\left[X_2, X_0\right]$$
$$\mathbb{P}^D\left[X_0\right] = \begin{pmatrix} \frac{1}{3} \\ \frac{2}{3} \end{pmatrix}\left[X_0\right]$$

In case $b)$ the optimal decorating conditional distributions would be

$$\mathbb{P}^D\left[X_0 | X_1\right] = \begin{pmatrix} \frac{1}{2} & \frac{1}{3} \\ \frac{1}{2} & \frac{2}{3} \end{pmatrix}\left[X_0, X_1\right]$$
$$\mathbb{P}^D\left[X_2 | X_1\right] = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}\left[X_2, X_1\right]$$
$$\mathbb{P}^D\left[X_1\right] = \begin{pmatrix} \frac{1}{3} \\ \frac{2}{3} \end{pmatrix}\left[X_1\right]$$

Task 4:

With respect to the empirical distribution with respect to the extended dataset no conditional independence assumption would hold. The Bayesian networks in task 3 would be the naive chain decompositions (see lecture II.1-Independence) of the empirical distributions.